Normal Learning Matrix and Course Matrix

Semester 1, Year 1

Course Code	Course Title	Core or Elective	Lectures	Seminars /Tutorials	Assignments	Independent Study	Practical	Total hours	Credits
CDF 111	Digital Forensics and Cybersecurity Essentials	Core	15	30	15	15	45	120	12
BIT 114	Computer Programming	Core	15	30	15	15	45	120	12
MTH 112	Discrete Mathematics	Core	45	15	15	15	0	90	9
BIT 112	Introduction to Computer Networking	Core	15	30	15	15	45	120	12
BIT 117	Computer Troubleshooting and Maintenance	Core	15	15	15	15	30	90	9
	Total of Core courses								54
SC 101	Communication Skills	Elective	30	15	15	15	15	90	9
BIT 113	Fundamentals of Digital Electronics	Elective	15	15	15	15	30	90	9
FIN 113	Financial Literacy	Elective						90	9
BIT 111	Microcomputers and Applications	Elective	15	30	15	15	45	120	12
	Total of Elective courses								39
	Total for Semester 1 year 1								93

Semester 2, Year1

Course Code	Course Title	Core or Elective	Lectures	Seminars /Tutorials	Assignments	Independent Study	Practical	Total hours	Credits
BIT 121	Operating Systems and Computer Architecture	Core	15	30	15	15	45	120	12
BIT 122	Data Structures and Algorithms	Core	15	30	15	15	45	120	12
BIT 126	Capstone Project I	Core	0	30		30	60	120	12
BIT 124	Programming in Python	Core	15	30	15	15	45	120	12
	Total of Core courses								48
BIT 127	Business Information Technology	Elective	15	30	15	15	45	120	12
BIT 125	Multimedia, Animation and Graphics	Elective	15	30	15	15	45	120	12
	Total of Elective courses								24
	Total for Semester 2 year 1								72

Course Description

i. **Course Title: BIT 111 – Microcomputers and Applications**

ii. **Course Aim**

This course aims to enhance knowledge and skills to students undertaking various fields to be able to apply computer technologies in their relevant field.

iii. **Course Expected Learning Outcomes**

Upon successful completion of this course the student will be expected to:

1. Identify the principal hardware components of a microcomputer and describe their functions.
2. Use an operating system to manage computer operations.
3. Apply basic personal computer (PC) security control.
4. Use word processing application software to perform basic tasks.
5. Use a spreadsheet application software to perform basic tasks.
6. Apply the knowledge of Computer applications to solve various real life problems.

iv. **Course Status**: Core

v. **Credits Rating**: 12

vi. **Total Hours Spent**: 120

vii. **Course Content**

1. Introduction (face-to-face)
   - 1.1. Computer Generations
   - 1.2. Computer history
   - 1.3. Computer Classifications (Classification according to size, technology, purpose, network roles, speed.
2. Main Components of a Microcomputer: (face-to-face)
   - 2.1. Devices and their logical configurations (including CPU, printers, disk drives, keyboards monitors, etc),
   - 2.2. Comparative aspects of micro and mainframe, workstations and micro, minicomputers and microcomputers.
3. PC Security Controls: (online)
   - 3.1. hard disk (power on) password,
   - 3.2. BIOS password,
   - 3.3. login password.
4. Functions and Types of Operating Systems: (online)
   - 4.1. (MS-DOS, UNIX/LINUX, XENIX, etc.);
   - 4.2. The Disk operating system (DOS and its functions,
   - 4.3. Windows and its applications.
5. Microcomputer Applications: (face-to-face)
   - 5.1. Word processors (e.g. Word): how to open and start a document, save and print a document and other basic tasks.

5.2. Spreadsheets: To make a simple spreadsheet (e.g. by using Excel), do some simple operations, saving and printing the sheet and other basic tasks.

5.3. Concepts of web browsing and communication using emails.

6. Data Security: (online)

6.1. General concepts of protecting data and information against physical destruction

6.2. Concepts of controlling unauthorized access.

6.3. Malware

viii. **Teaching and Learning Activities**

Teaching and learning activities will include interactive online and face-to-face lectures, class discussions and brainstorming to highlight everyday computer applications and examples, brainstorming relevant uses of computers, demonstrate features and cases of using Word processor and Spreadsheet, students practice using Word processor and Spreadsheet, as well as install anti-malware and scan computers.

ix. **Assessment Methods**

Theory 20% (Exercises in form of quizzes, presentation of scenarios to demonstrate understanding of concepts computers and applications, independent and group Word Processor and Spreadsheet program use assignments, theory test).

Practical 40% (Practical hands on test (using Word Processor and Spreadsheet programs) to examine students ability to use Word and Spreadsheet programs), practical report. End of semester university examination 40%.

x. **Readings List**

1. Shelly, G. A. (2011). *Microsoft Office 2010 Introductory,* Boston: Cengage Learning
2. Behrooz P. (2005). *Computer architecture: from microprocessors to supercomputers,* New York: Oxford University Press, 2005
3. Margaret Levine Young, Alison Barrows, and Joseph C Stockman (2010). *Access 2010 ALL-IN-ONE For Dummies:* Wiley Publishing Inc.
4. Howard W. P. (2004). *An introduction to microcomputer applications packages for university faculty,* THE Journal (Technological Horizons In Education)
5. Gammill, R. C. (1982). Microcomputers: Applications, Problems, and Promises. *Advances in Computers*, 155–224. https://doi.org/10.1016/s0065-2458(08)60569-0.
6. Castellan, N. J. (1988, March). Comments on applications of microcomputers in teaching. *Behavior Research Methods, Instruments, & Computers, 20*(2), 193–196. https://doi.org/10.3758/bf03203830.